



BTS-IV-04.19-0652

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B.Tech. Degree IV Semester Examination April 2019

ME 15-1402 METROLOGY AND INSTRUMENTATION (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain the term cosine error with suitable sketch.
(b) State the disadvantages of material standard.
(c) Define least count of vernier instrument. How is it determined?
(d) Explain the principle of auto collimator with a neat sketch.
(e) Describe briefly the information needed for designating surface finish on drawings, with an example.
(f) Differentiate between static characteristics and dynamic characteristics of an instrument.
(g) Distinguish between active and passive transducers.
(h) What are the errors in strain gauge? Explain.
(i) Define load cells. What are the types of load cells?
(j) What are the major differences between a thermocouple and an RTD?

PART B

(4 × 10 = 40)

- II. (a) Differentiate between hole basis system and shaft basis system. (4)
(b) What are the different classification of fits? Explain with the help of neat sketches. (6)
- OR**
- III. (a) Discuss about the importance of gauging in manufacturing processes. With neat sketches, explain the different classification of gauges. (6)
(b) Explain what is meant by interchangeability and selective assembly. (4)
- IV. (a) Draw a neat sketch to illustrate the use of sine bar for measurement of taper plug gauge and explain it briefly. (5)
(b) What is a clinometer? Describe how can it be used for measurement and setting of angles. Illustrate your answer with sketches. (5)
- OR**
- V. (a) Explain the working of laser beam interferometer with the help of a neat sketch. (7)
(b) Describe the precautions to be taken while using optical flat. (3)

(P.T.O.)

- VI. (a) What are the main functional elements of an instrument? Mention the functions of each element. (6)
- (b) Define the following terms : (4)
- (i) Speed of response
 - (ii) Fidelity and dynamic error
- OR**
- VII. What are the different methods of correcting spurious inputs in an instrument? Explain with examples. (10)
- VIII. (a) Explain the working of ORSAT apparatus. (5)
- (b) Explain the construction and working of Geiger Muller counter. (5)
- OR**
- IX. (a) Explain with a neat sketch, weight measurement using multiple lever system. (6)
- (b) Differentiate between unbonded and bonded strain gauges. (4)
